

UDF

Reading: User-Defined Schema (UDS) for DSL and SQL

Estimated time needed: 10 minutes

How to Define and Enforce a User-Defined Schema in PySpark?

In this reading, you will learn how to define and enforce a user-defined schema in PySpark.

Spark provides a structured data processing framework that can define and enforce schemas for various data sources, including CSV files. Let's look at the steps to define and use a user-defined schema for a CSV file in PySpark:

Step 1:

Import the required libraries.

```
1 from pyspark.sql.types import StructType, IntegerType, FloatType, StringType, StructField
```

Step 2:

Define the schema.

Understanding the data before defining a schema is an important step.

Let's take a look at the step-by-step approach to understanding the data and defining an appropriate schema for a given input file:

1. **Explore the data:** Understand the different data types present in each column.
2. **Column data types:** Determine the appropriate data types for each column based on your observed values.
3. **Define the schema:** Use the 'StructType' class in Spark and create a 'StructField' for each column, mentioning the column name, data type, and other properties.

Example:

```
1 schema = StructType([
2     StructField("Emp_Id", StringType(), False),
3     StructField("Emp_Name", StringType(), False),
4     StructField("Department", StringType(), False),
5     StructField("Salary", IntegerType(), False),
6     StructField("Phone", IntegerType(), True),
7 ])
```

'False' indicates null values are NOT allowed for the column.

The schema defined above can be utilized for the below CSV file data:

Filename: employee.csv

```
1 emp_id,emp_name,dept,salary,phone
2 A101,jhon,computer science,1000,+1 (701) 846 958
3 A102,Peter,Electronics,2000,
4 A103,Micheal,IT,2500,
```

Step 3: Read the input file with user-defined schema.

```
1 #create a dataframe on top a csv file
2 df = (spark.read
3     .format("csv")
4     .schema(schema)
5     .option("header", "true")
6     .load("employee.csv")
7 )
8 # display the dataframe content
9 df.show()
```

Step 4: Use the `printSchema()` method in Spark to display the schema of a DataFrame and ensure that the schema is applied correctly to the data.

```
1 df.printSchema()
```

Through the preceding four steps, you've acquired the ability to establish a schema for a CSV file. Additionally, you've employed this user-defined schema (UDF) to read the CSV file, exhibit its contents, and showcase the schema itself.